LAB 3: Student Startup Crowdfunding Platform

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1. Architectural Choice

The system uses a Component-Based Architecture in which the main functionality is divided into separate components: Campaign Management, Pledge & Payment Service, and Milestone Verification. Client interfaces, the external payment gateway, and the crowdfunding database communicate with these components through defined ports and interfaces.

2. Two Reasons for the Choice

- Modular functionality: campaign creation and milestone definition, backer pledges and payments, and milestone verification are separated into independent components. This makes the platform easier to develop and maintain.
- Clear communication between modules: the components exchange information such as Campaign Requests, Pledge/Reward Requests, MilestoneReviews, Campaign Data, Payment Data, and Verification Data through ports/interfaces. This reduces direct dependency between internal modules and supports easier modification.

3. Security Advantage

The architecture exposes functionality through defined interfaces rather than giving every component direct access to another component's internal operations. Payment processing is isolated through the Payment Gateway interface, while milestone verification is exposed through a dedicated verification interface. This supports controlled access and reduces unnecessary exposure between components.

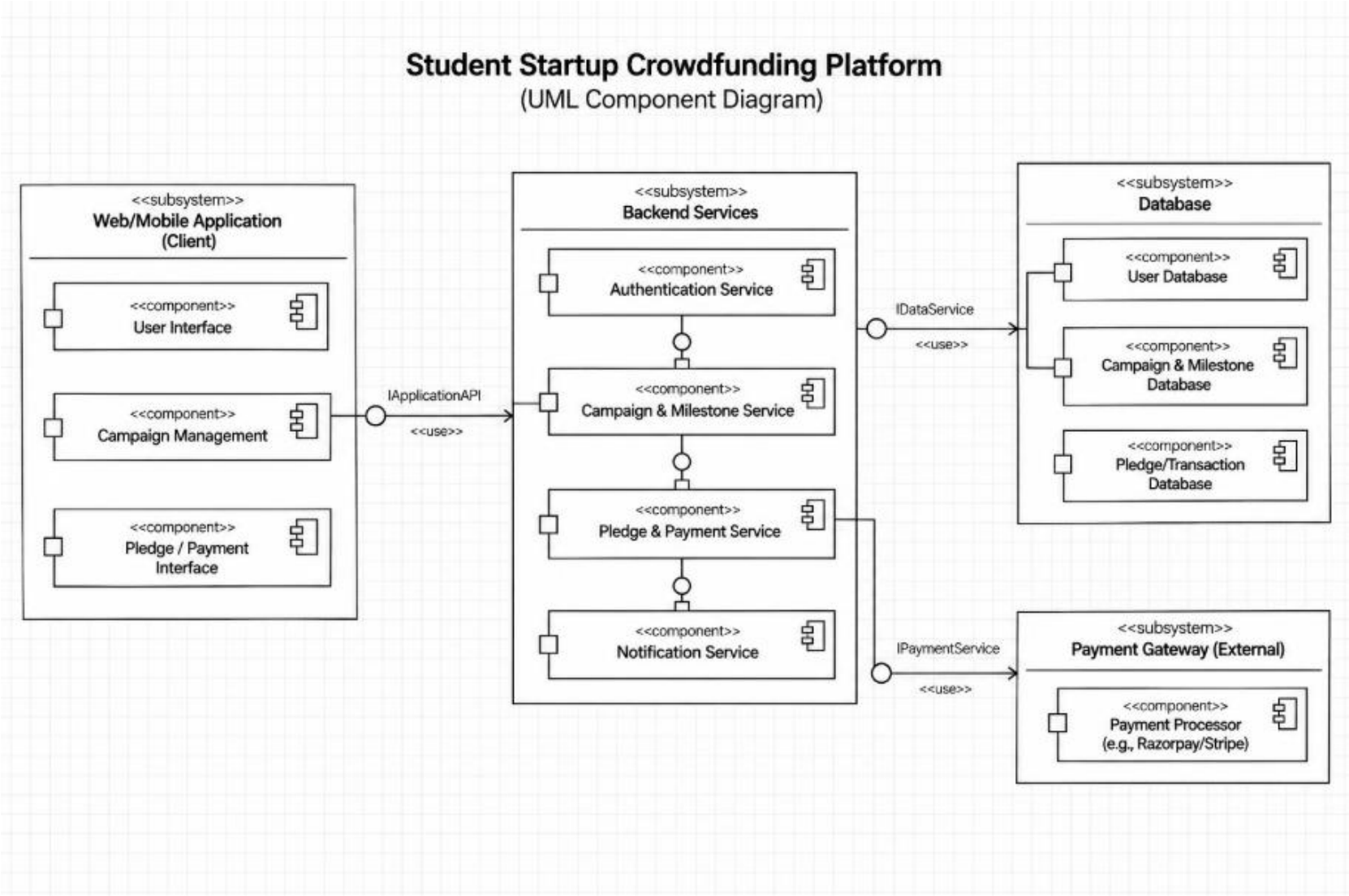
4. Performance Benefit

Independent components can handle different operations without requiring the entire platform to be processed as one unit. For example, campaign browsing and management, pledge/payment processing, and milestone verification can operate independently when their required data is available. This reduces unnecessary processing and can improve responsiveness.

5. Component Diagram

The next page shows the component-based architecture using ports, provided interfaces, required interfaces, and connectors.

5. Component Diagram —



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